

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

In [2]: data = pd.read_csv('US_Accidents_March23_sampled_500k.csv')
data.head(10)
```

	ID	Source	Severity	Start_Time	End_Time	Start_Lat	Start_Lng	End_Lat	End_Lng	Distance(mi)	...	Roundabout	Station	Stop	Traffic_Calming	Traffic_Signal	Sunrise	Sunset
0	A-2047758	Source2	2	2019-06-12 10:10:56	2019-06-12 10:55:58	30.641211	-91.153481	NaN	NaN	0.000	...	False	False	False	False	False	True	False
1	A-4694324	Source1	2	2022-12-03 23:37:14.0000000000	2022-12-04 01:56:53.0000000000	38.990562	-77.399070	38.990037	-77.398282	0.056	...	False	False	False	False	False	True	False
2	A-5006183	Source1	2	2022-08-20 13:13:00.0000000000	2022-08-20 15:22:45.0000000000	34.661189	-120.492822	34.661189	-120.492442	0.022	...	False	False	False	False	False	True	False
3	A-4237356	Source1	2	2022-02-21 17:43:04	2022-02-21 19:43:23	43.680592	-92.993317	43.680574	-92.972223	1.054	...	False	False	False	False	False	True	False
4	A-6690583	Source1	2	2020-12-04 01:46:00	2020-12-04 04:13:09	35.395484	-118.985176	35.395476	-118.985995	0.046	...	False	False	False	False	False	True	False
5	A-1101469	Source2	2	2021-03-29 07:03:58	2021-03-29 08:51:01	42.532082	-70.944267	NaN	NaN	0.000	...	False	False	False	False	False	True	False
6	A-7222249	Source1	2	2020-01-14 16:49:23	2020-01-14 20:49:23	42.421280	-123.119450	42.421280	-123.119450	0.000	...	False	False	False	False	False	True	False
7	A-6188239	Source1	2	16:48:00.0000000000	19:09:09.0000000000	30.191010	-85.682508	30.190329	-85.682504	0.047	...	False	False	False	False	False	True	False
8	A-4222549	Source1	2	2022-10-12 13:59:30	2022-10-12 15:33:53	32.868947	-96.804018	32.868950	-96.804010	0.038	...	False	True	False	False	False	True	False
9	A-5924038	Source1	2	2021-10-21 07:39:30	2021-10-21 09:24:30	39.717218	-86.124691	39.733478	-86.137021	1.301	...	False	False	False	False	False	True	False

10 rows x 46 columns

```
In [3]: # Quick overview
df=data
print(df.info())

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500000 entries, 0 to 499999
Data columns (total 46 columns):
 #   Column              Non-Null Count  Dtype
---  --
 0   ID                   500000 non-null object
 1   Source               500000 non-null object
 2   Severity             500000 non-null int64
 3   Start_Time           500000 non-null object
 4   End_Time             500000 non-null object
 5   Start_Lat            500000 non-null float64
 6   Start_Lng            500000 non-null float64
 7   End_Lat              279523 non-null float64
 8   End_Lng              279623 non-null float64
 9   Distance(mi)         500000 non-null float64
10   Description          499999 non-null object
11   Street              499389 non-null object
12   City                 499981 non-null object
13   County              500000 non-null object
14   State               500000 non-null object
15   Zipcode             499884 non-null object
16   Country             500000 non-null object
17   Timezone            499493 non-null object
18   Airport_Code        498554 non-null object
19   Weather_Timestamp   492326 non-null object
20   Temperature(F)      489534 non-null float64
21   Wind_Chill(F)       378983 non-null float64
22   Humidity(%)         488870 non-null float64
23   Pressure(in)        491072 non-null float64
24   Visibility(mi)      488709 non-null float64
25   Wind_Direction      488983 non-null object
26   Wind_Speed(mph)     463813 non-null float64
27   Precipitation(in)   357384 non-null float64
28   Weather_Condition   489539 non-null object
29   Amenity             500000 non-null bool
30   Bump               500000 non-null bool
31   Crossing            500000 non-null bool
32   Give_Way           500000 non-null bool
33   Junction            500000 non-null bool
34   No_Exit            500000 non-null bool
35   Railway            500000 non-null bool
36   Roundabout         500000 non-null bool
37   Station            500000 non-null bool
38   Stop               500000 non-null bool
39   Traffic_Calming    500000 non-null bool
40   Traffic_Signal     500000 non-null bool
41   Turning_Loop       500000 non-null bool
42   Sunrise_Sunset     498517 non-null object
43   Civil_Twilight     498517 non-null object
44   Nautical_Twilight  498517 non-null object
45   Astronomical_Twilight 498517 non-null object
dtypes: bool(13), float64(12), int64(1), object(20)
memory usage: 132.1+ MB
None
```

```
In [4]: # Count missing values for each column
missing_values = df.isnull().sum()
print('Missing Values:\n', missing_values[missing_values > 0])

# Percentage of missing values
missing_percentage = (df.isnull().sum() / len(df)) * 100
print('Missing Percentage:\n', missing_percentage[missing_percentage > 0])

Missing Values:
End_Lat      228377
End_Lng      228377
Description    1
Street        691
City          19
Zipcode       116
Timezone      587
Airport_Code  1446
Weather_Timestamp 7674
Temperature(F) 10466
Wind_Chill(F) 129017
Humidity(%)   11130
Pressure(in)   8928
Visibility(mi) 11291
Wind_Direction 11197
Wind_Speed(mph) 36987
Precipitation(in) 142616
Weather_Condition 11101
Sunrise_Sunset 1483
Civil_Twilight 1483
Nautical_Twilight 1483
Astronomical_Twilight 1483
dtype: int64

Missing Percentage:
End_Lat      44.0754
End_Lng      44.0754
Description    0.0002
Street        0.1382
City          0.0038
Zipcode       0.0232
Timezone      0.1014
Airport_Code  0.2892
Weather_Timestamp 1.5348
Temperature(F) 2.0932
Wind_Chill(F) 25.8034
Humidity(%)   2.2260
Pressure(in)   1.7856
Visibility(mi) 2.2582
Wind_Direction 2.2394
Wind_Speed(mph) 7.3974
Precipitation(in) 28.5232
Weather_Condition 2.2202
Sunrise_Sunset 0.2966
Civil_Twilight 0.2966
Nautical_Twilight 0.2966
Astronomical_Twilight 0.2966
dtype: float64
```

```
In [5]: # Count unique values per column
unique_values = df.nunique()
print("\nUnique Values:\n", unique_values)

Unique Values:
ID              500000
Source          3
Severity         4
Start_Time     487027
End_Time       493821
Start_Lat      368965
Start_Lng      378195
End_Lat        228333
End_Lng        221463
Distance(mi)   18627
Description     408146
Street         88272
City           9408
County         1686
State          49
Zipcode        127804
Country         1
Timezone        4
Airport_Code    1878
Weather_Timestamp 255623
Temperature(F)  712
Wind_Chill(F)  797
Humidity(%)     100
Pressure(in)     908
Visibility(mi)   65
Wind_Direction  24
Wind_Speed(mph) 95
Precipitation(in) 175
Weather_Condition 108
Amenity          2
Bump             2
Crossing         2
Give_Way         2
Junction         2
No_Exit          2
Railway          2
Roundabout      2
Station          2
Stop             2
Traffic_Calming 2
Traffic_Signal   2
Turning_Loop     1
Sunrise_Sunset  1
Civil_Twilight  2
Nautical_Twilight 2
Astronomical_Twilight 2
dtype: int64
```

```
In [6]: # Drop high-missing columns
threshold = 0.4 # Drop columns with more than 40% missing values
missing_percentage = df.isnull().mean()
high_missing_cols = missing_percentage[missing_percentage > threshold].index
print(f"ndropping columns with >{threshold} % missing values: {list(high_missing_cols)}")
df.drop(columns=high_missing_cols, inplace=True)

Dropping columns with >40.0% missing values: ['End_Lat', 'End_Lng']
```

```
In [7]: # Impute numerical columns with median
numerical_cols = df.select_dtypes(include=['float64', 'int64']).columns
for col in numerical_cols:
    if df[col].isnull().sum() > 0:
        df[col].fillna(df[col].median(), inplace=True)
        print(f"Imputed missing values in {col} with median.")

Imputed missing values in Temperature(F) with median.
Imputed missing values in Wind_Chill(F) with median.
Imputed missing values in Humidity(%) with median.
Imputed missing values in Pressure(in) with median.
Imputed missing values in Visibility(mi) with median.
Imputed missing values in Wind_Speed(mph) with median.
Imputed missing values in Precipitation(in) with median.
```

```
In [8]: # Impute categorical columns with mode
categorical_cols = df.select_dtypes(include=['object']).columns
for col in categorical_cols:
    if df[col].isnull().sum() > 0:
        df[col].fillna(df[col].mode()[0], inplace=True)
        print(f"Imputed missing values in {col} with mode.")

Imputed missing values in Description with mode.
Imputed missing values in Street with mode.
Imputed missing values in City with mode.
Imputed missing values in Zipcode with mode.
Imputed missing values in Timezone with mode.
Imputed missing values in Airport_Code with mode.
Imputed missing values in Weather_Timestamp with mode.
Imputed missing values in Wind_Direction with mode.
Imputed missing values in Weather_Condition with mode.
Imputed missing values in Sunrise_Sunset with mode.
Imputed missing values in Civil_Twilight with mode.
Imputed missing values in Nautical_Twilight with mode.
Imputed missing values in Astronomical_Twilight with mode.
```

```
In [9]: # Drop low-variance columns
low_variance_cols = [col for col in df.columns if df[col].nunique() == 1]
print(f"ndropping low-variance columns: {low_variance_cols}")
df.drop(columns=low_variance_cols, inplace=True)

Dropping low-variance columns: ['Country', 'Turning_Loop']
```

```
In [10]: data['Start_Time'] = pd.to_datetime(data['Start_Time'], errors='coerce')
data['End_Time'] = pd.to_datetime(data['End_Time'], errors='coerce')
data['Accident_Duration'] = df['End_Time'] - df['Start_Time']
).dt.total_seconds() / 3600 # Convert to hours
```

```
In [11]: # Simplify 'Weather_Condition' by grouping similar terms
weather_mapping = {
    'Clear': 'Clear',
    'Sunny': 'Clear',
    'Cloudy': 'Cloudy',
    'Overcast': 'Cloudy',
    # Add other mappings as needed
}

df['Weather_Condition'] = df['Weather_Condition'].map(weather_mapping).fillna('Other')

# Simplify 'Wind_Direction'
df['Wind_Direction'] = df['Wind_Direction'].apply(
    lambda x: 'Caln' if x in ['Caln', 'Variable'] else x
)

In [12]: # Validate the cleaned dataset
print("\nDataset Info After Cleaning:")
print(df.info())

# Check for remaining missing values
print("\nRemaining Missing Values:")
print(df.isnull().sum() if df.isnull().sum() > 0)

# Save the cleaned dataset
output_file = 'cleaned_dataset.csv'
df.to_csv(output_file, index=False)
print(f"\nCleaned dataset saved to {output_file}")

Dataset Info After Cleaning:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500000 entries, 0 to 499999
Data columns (total 43 columns):
 #   Column              Non-Null Count  Dtype
---  --
 0   ID                   500000 non-null object
 1   Source               500000 non-null object
 2   Severity             500000 non-null int64
 3   Start_Time           451837 non-null datetime64[ns]
 4   End_Time             451837 non-null datetime64[ns]
 5   Start_Lat            500000 non-null float64
 6   Start_Lng            500000 non-null float64
 7   Distance(mi)         500000 non-null float64
 8   Description          500000 non-null object
 9   Street              500000 non-null object
10   City                 500000 non-null object
11   County              500000 non-null object
12   State               500000 non-null object
13   Zipcode             500000 non-null object
14   Timezone            500000 non-null object
15   Airport_Code        500000 non-null object
16   Weather_Timestamp   500000 non-null object
17   Temperature(F)      500000 non-null float64
18   Wind_Chill(F)       500000 non-null float64
19   Humidity(%)         500000 non-null float64
20   Pressure(in)        500000 non-null float64
21   Visibility(mi)      500000 non-null float64
22   Wind_Direction      500000 non-null object
23   Wind_Speed(mph)     500000 non-null float64
24   Precipitation(in)   500000 non-null float64
25   Weather_Condition   500000 non-null object
26   Amenity             500000 non-null bool
27   Bump               500000 non-null bool
28   Crossing            500000 non-null bool
29   Give_Way           500000 non-null bool
30   Junction            500000 non-null bool
31   No_Exit            500000 non-null bool
32   Railway            500000 non-null bool
33   Roundabout         500000 non-null bool
34   Station            500000 non-null bool
35   Stop               500000 non-null bool
36   Traffic_Calming    500000 non-null bool
37   Traffic_Signal     500000 non-null bool
38   Sunrise_Sunset     500000 non-null object
39   Civil_Twilight     500000 non-null object
40   Nautical_Twilight  500000 non-null object
41   Astronomical_Twilight 500000 non-null object
42   Accident_Duration   451837 non-null float64
dtypes: bool(12), datetime64[ns](2), float64(11), int64(1), object(17)
memory usage: 124.0+ MB
None

Remaining Missing Values:
Start_Time      48163
End_Time        48163
Accident_Duration 48163
dtype: int64

Cleaned dataset saved to cleaned_dataset.csv
```

```
In [13]: # Option 1: Fill missing values in 'Accident_Duration' with the median
data['Accident_Duration'].fillna(data['Accident_Duration'].median(), inplace=True)

# Drop rows where either 'Start_Time' or 'End_Time' is missing
data.dropna(subset=['Start_Time', 'End_Time'], inplace=True)

# Check the remaining missing values
print(data.isna().sum())

ID              0
Source          0
Severity        0
Start_Time      0
End_Time        0
Start_Lat       0
Start_Lng       0
Distance(mi)    0
Description      0
Street          0
County          0
City            0
State           0
Zipcode         0
Timezone        0
Airport_Code    0
Weather_Timestamp 0
Temperature(F)  0
Wind_Chill(F)  0
Humidity(%)     0
Pressure(in)    0
Visibility(mi)   0
Wind_Direction  0
Wind_Speed(mph) 0
Precipitation(in) 0
Weather_Condition 0
Amenity         0
Bump            0
Crossing        0
Give_Way        0
Junction        0
No_Exit         0
Railway         0
Roundabout      0
Station         0
Stop            0
Traffic_Calming 0
Traffic_Signal  0
Sunrise_Sunset 0
Civil_Twilight  0
Nautical_Twilight 0
Astronomical_Twilight 0
Accident_Duration 0
dtype: int64
```

```
In [14]: df.head()
```

	ID	Source	Severity	Start_Time	End_Time	Start_Lat	Start_Lng	Distance(mi)	Description	Street	...	Roundabout	Station	Stop	Traffic_Calming	Traffic_Signal	Sunrise	Sunset
0	A-2047758	Source2	2	2019-06-12 10:10:56	2019-06-12 10:55:58	30.641211	-91.153481	0.000	Accident on LA-19 Baker-Zachary Hwy at Lower ...	Highway 19	...	False	False	False	False	False	True	False
3	A-4237356	Source1	2	2022-02-21 17:43:04	2022-02-21 19:43:23	43.680592	-92.993317	1.054	Incident on I-90 EB near REST AREA Drive with ...	14th St NW	...	False	False	False	False	False	True	False
4	A-6690583	Source1	2	2020-12-04 01:46:00	2020-12-04 04:13:09	35.395484	-118.985176	0.046	RP ADV THEY LOCATED SUSP VEH OF 20002 - 726 CR...	River Blvd	...	False	False	False	False	False	True	False
5	A-1101469	Source2	2	2021-03-29 07:03:58	2021-03-29 08:51:01	42.532082	-70.944267	0.000	Accident on Forest St at Lowell St.	Lowell St	...	False	False	False	False	False	True	False
6	A-7222249	Source1	2	2020-01-14 16:49:23	2020-01-14 20:49:23	42.421280	-123.119450	0.000	AT OR-99 EB at 45A - Accident.	I-5 N	...	False	False	False	False	False	True	False

5 rows x 43 columns

```
In [31]: # Option 1: Fill missing values in 'Accident_Duration' with the median
data['Accident_Duration'].fillna(data['Accident_Duration'].median(), inplace=True)

# Drop rows where either 'Start_Time' or 'End_Time' is missing
data.dropna(subset=['Start_Time', 'End_Time'], inplace=True)

# Check the remaining missing values
print(data.isna().sum())

ID              0
Source          0
Severity        0
Start_Time      0
End_Time        0
Start_Lat       0
Start_Lng       0
Distance(mi)    0
Description      0
Street          0
County          0
City            0
State           0
Zipcode         0
Timezone        0
Airport_Code    0
Weather_Timestamp 0
Temperature(F)  0
Wind_Chill(F)  0
Humidity(%)     0
Pressure(in)    0
Visibility(mi)   0
Wind_Direction  0
Wind_Speed(mph) 0
Precipitation(in) 0
Weather_Condition 0
Amenity         0
Bump            0
Crossing        0
Give_Way        0
Junction        0
No_Exit         0
Railway         0
Roundabout      0
Station         0
Stop            0
Traffic_Calming 0
Traffic_Signal  0
Sunrise_Sunset 0
Civil_Twilight  0
Nautical_Twilight 0
Astronomical_Twilight 0
Accident_Duration 0
dtype: int64
```

```
In [14]: df.head()
```

	ID	Source	Severity	Start_Time	End_Time	Start_Lat	Start_Lng	Distance(mi)	Description	Street	...	Roundabout	Station	Stop	Traffic_Calming	Traffic_Signal	Sunrise	Sunset
0	A-2047758	Source2	2	2019-06-12 10:10:56	2019-06-12 10:55:58	30.641211	-91.153481	0.000	Accident on LA-19 Baker-Zachary Hwy at Lower ...	Highway 19	...	False	False	False	False	False	True	False
3	A-4237356	Source1	2	2022-02-21 17:43:04	2022-02-21 19:43:23	43.680592	-92.993317	1.054	Incident on I-90 EB near REST AREA Drive with ...	14th St NW	...	False	False	False	False	False	True	False
4	A-6690583	Source1	2	2020-12-04 01:46:00	2020-12-04 04:13:09	35.395484	-118.985176	0.046	RP ADV THEY LOCATED SUSP VEH OF 20002 - 726 CR...	River Blvd	...	False	False	False	False	False	True	False
5	A-1101469	Source2	2	2021-03-29 07:03:58	2021-03-29 08:51:01	42.532082	-70.944267	0.000	Accident on Forest St at Lowell St.	Lowell St	...	False	False	False	False	False	True	False
6	A-7222249	Source1	2	2020-01-14 16:49:23	2020-01-14 20:49:23	42.421280	-123.119450	0.000	AT OR-99 EB at 45A - Accident.	I-5 N	...	False	False	False	False	False	True	False

5 rows x 43 columns

```
In [31]: # Option 1: Fill missing values in 'Accident_Duration' with the median
data['Accident_Duration'].fillna(data['Accident_Duration'].median(), inplace=True)

# Drop rows where either 'Start_Time' or 'End_Time' is missing
data.dropna(subset=['Start_Time', 'End_Time'], inplace=True)

# Check the remaining missing values
print(data.isna().sum())

ID              0
Source          0
Severity        0
Start_Time      0
End_Time        0
Start_Lat       0
Start_Lng       0
Distance(mi)    0
Description      0
Street          0
County          0
City            0
State           0
Zipcode         0
Timezone        0
Airport_Code    0
Weather_Timestamp 0
Temperature(F)  0
Wind_Chill(F)  0
Humidity(%)     0
Pressure(in)    0
Visibility(mi)   0
Wind_Direction  0
Wind_Speed(mph) 0
Precipitation(in) 0
Weather_Condition 0
Amenity         0
Bump            0
Crossing        0
Give_Way        0
Junction        0
No_Exit         0
Railway         0
Roundabout      0
Station         0
Stop            0
Traffic_Calming 0
Traffic_Signal  0
Sunrise_Sunset 0
Civil_Twilight  0
Nautical_Twilight 0
Astronomical_Twilight 0
Accident_Duration 0
dtype: int64
```

```
In [14]: df.head()
```

	ID	Source	Severity	Start_Time	End_Time	Start_Lat	Start_Lng	Distance(mi)	Description	Street	...	Roundabout	Station	Stop	Traffic_Calming	Traffic_Signal	Sunrise	Sunset
0	A-2047758	Source2	2	2019-06-12 10:10:56	2019-06-12 10:55:58	30.641211	-91.153481	0.000	Accident on LA-19 Baker-Zachary Hwy at Lower ...	Highway 19	...	False	False	False	False	False	True	False
3	A-4237356	Source1	2	2022-02-21 17:43:04	2022-02-21 19:43:23	43.680592	-92.993317	1.054	Incident on I-90 EB near REST AREA Drive with ...	14th St NW	...	False	False	False	False	False	True	False
4	A-6690583	Source1	2	2020-12-04 01:46:00	2020-12-04 04:13:09	35.395484	-118.985176	0.046	RP ADV THEY LOCATED SUSP VEH OF 20002 - 726 CR...	River Blvd	...	False	False	False	False	False	True	False
5	A-1101469	Source2	2	2021-03-29 07:03:58	2021-03-29 08:51:01	42.532082	-70.944267	0.000	Accident on Forest St at Lowell St.	Lowell St	...	False	False	False	False	False	True	False
6	A-7222249	Source1	2	2020-01-14 16:49:23	2020-01-14 20:49:23	42.421280	-123.119450	0.000	AT OR-99 EB at 45A - Accident.	I-5 N	...	False	False	False	False	False	True	False

5 rows x 43 columns

```
In [31]: # Option 1: Fill missing values in 'Accident_Duration' with the median
data['Accident_Duration'].fillna(data['Accident_Duration'].median(), inplace=True)

# Drop rows where either 'Start_Time' or 'End_Time' is missing
data.dropna(subset=['Start_Time', 'End_Time'], inplace=True)

# Check the remaining missing values
print(data.isna().sum())

ID              0
Source          0
Severity        0
Start_Time      0
End_Time        0
Start_Lat       0
Start_Lng       0
Distance(mi)    0
Description      0
Street          0
County          0
City            0
State           0
Zipcode         0
Timezone        0
Airport_Code    0
Weather_Timestamp 0
Temperature(F)  0
Wind_Chill(F)  0
Humidity(%)     0
Pressure(in)    0
Visibility(mi)   0
Wind_Direction  0
Wind_Speed(mph) 0
Precipitation(in) 0
Weather_Condition 0
Amenity         0
Bump            0
Crossing        0
Give_Way        0
Junction        0
No_Exit         0
Railway         0
Roundabout      0
Station         0
Stop            0
Traffic_Calming 0
Traffic_Signal  0
Sunrise_Sunset 0
Civil_Twilight  0
Nautical_Twilight 0
Astronomical_Twilight 0
Accident_Duration 0
dtype: int64
```

```
In [14]: df.head()
```

	ID	Source	Severity	Start_Time	End_Time	Start_Lat	Start_Lng	Distance(mi)	Description	Street	...	Roundabout	Station	Stop	Traffic_Calming	Traffic_Signal	Sunrise	Sunset
0	A-2047758	Source2	2	2019-06-12 10:10:56	2019-06-12 10:55:58	30.641211	-91.153481	0.000	Accident on LA-19 Baker-Zachary Hwy at Lower ...	Highway 19	...	False	False	False	False	False	True	False
3	A-4237356	Source1	2	2022-02-21 17:43:04	2022-02-21 19:43:23	43.680592	-92.993317	1.054	Incident on I-90 EB near REST AREA Drive with ...	14th St NW	...	False	False	False	False	False	True	False
4	A-6690583	Source1	2	2020-12-04 01:46:00	2020-12-04 04:13:09	35.395484	-118.985176	0.046	RP ADV THEY LOCATED SUSP VEH OF 20002 - 726 CR...	River Blvd	...	False	False	False	False	False	True	False
5	A-1101469	Source2	2	2021-03-29 07:03:58	2021-03-29 08:51:01	42.532082	-70.944267	0.000	Accident on Forest St at Lowell St.	Lowell St	...	False	False	False	False	False	True	False
6	A-7222249	Source1	2	2020-01-14 16:49:23	2020-01-14 20:49:23	42.421280	-123.119450	0.000	AT OR-99 EB at 45A - Accident.	I-5 N	...	False	False	False	False	False	True	False

5 rows x 43 columns

```
In [31]: # Option 1: Fill missing values in 'Accident_Duration' with the median
data['Accident_Duration'].fillna(data['Accident_Duration'].median(), inplace=True)

# Drop rows where either 'Start_Time' or 'End_Time' is missing
data.dropna(subset=['Start_Time', 'End_Time'], inplace=True)

# Check the remaining missing values
print(data.isna().sum())

ID              0
Source          0
Severity        0
Start_Time      0
End_Time       
```